Saitama AB
Lindhagensgatan 59
112 43 Stockholm
+46 (0)72 152 42 28
samuel.gabrielsson@gmail.com
samuelgabrielsson
proximus

Samuel Gabrielsson

Om Mig

Med över 15 års erfarenhet inom områden som **Senior Linux Embedded och Kernelutvecklare**, **Scrum Master**, **Produktägare** samt specialist i **systemintegration**, **IT-automation**, **testramverk-arkitektur** och **testfall**, kombinerar jag teknisk expertis med starkt ledarskap. Ett av mina mest stolta bidrag var att medverka i **NASA:s Ingenuity Mars-helikopter**, vilket möjliggjorde den första kontrollerade motorflygningen på en annan planet.

Jag är en problemlösare som närmar mig varje utmaning med nyfikenhet och uthållighet, och jag värdesätter samarbete och en positiv arbetsmiljö. Genom pålitlighet, öppenhet och kunskapsdelning bygger jag förtroende hos både kollegor och kunder.

Erfarenhet

Mar 14, 2024–
Mar 01, 2026

Utvecklare, Scrum Master och Produktägare, *Saitama AB*, Stockholm, Sweden

- **Konsult på Ericsson AB - Radio Software**
  Ett tvåårigt kontrakt inom RadioSW RX/Uplink-teamet, där jag balanserade tre roller samtidigt: utvecklare, Scrum Master och Product Owner för Massive MIMO Granada-programmet samt BALI-projektet för radioprodukten AAS AIR 6494.
  - Utvecklare under återstående tid, med fokus på Uplink/Rx-delen av mjukvaran, bestående av drivrutiner och testfall.
  - Scrum Master för Uplink/Rx-teamet. Driva arbetssättet (Way of Working) genom dagliga stand-ups, sprintgranskningar, retrospektiv, refinement och sprintplanering.
  - Productägare för Uplink/Rx-delen av produkten. Initial skapare av backloggen för hela RadioSW BALI-projektet i Jira. Skapa, förfina och planera backloggen för teamet.
  - Använda verktyg som Yocto, Jira, Git, Gerrit, Jenkins, C/C++, Java och NeoVim med AI-integration (Codeium).

Jun 01, 2023–
Dec 31, 2023

Embedded Developer och Scrum Master, *Saitama AB*, Stockholm, Sweden

- **Konsult på CrossControl AB - Uppsala**
  Scrum master och Linux Embedded Utvecklare. Ansvarade för två Scrumteam (ett Windows-team och ett Linux-team) samt utvecklade drivrutiner och applikationer för produkter som CCPilot X1200 och X900. CCPilot X1200 är en 12-tums display med prestanda i nivå med en bärbar dator för industriella fordon, utformad för att uppfylla krävande miljökrav. Den stöder moderna och nästa generations verktyg som sömlöst kan integrera flera kameraflöden i ditt HMI-system, vilket förbättrar operatörens sikt och säkerhet.
  - Leda teammöten, dagliga stand-ups, sprintgranskningar, retrospektiv, refinement och sprintplanering. Hålla backloggen uppdaterad med tillräcklig kvalitet. Skapa väl dokumenterade uppgifter och fördela dem med ett bra arbetsflöde.
  - Utveckla Linux kernel drivrutiner och in-house mjukvara för att stödja ljussensorn, EEPROM, automatisk bakgrundsbelysning, knappsatser, audio codec, PXE bootable installationsavbilder för kund och produktion, I2C-kommunikation samt CAN-FD MCU.
  - Använda verktyg som Yocto, Devtool, Jira, Git, Bitbucket, Jenkins samt C/C++, Python, Bash och Vim.

Okt 01, 2022–
Mar 31, 2023

Digital Radio Algorithms Developer, *Saitama AB*, Stockholm, Sweden

- **Konsult på Ericsson AB - Hårdvaruavdelningen**
  Pre-development av firmware för algoritmstyrning och funktioner i mikrokontrollers (även kallade Device Processors (DP) i Ericssons terminologi). Med hjälp av DP:er kan Power Amplifiers (effektförstärkarna) göras mer effektiva genom att minska den distorsion som uppstår i deras icke-linjära områden. Resultatet av uppdraget används sen av framtida radio produkter inom 6G.
  - Utveckla Digital Pre-Distortion (DPD)-mjukvara i Matlab och C för att öka effektiviteten hos effektförstärkare.
  - Använda Xilinx SDK eller Vitis IDE för att implementera och felsöka FPGA samt mjukvaruladdning på målplattformen.

Okt 01, 2018–
Sep 30, 2022

Senior Research Software Engineer, *Saitama AB*, Stockholm, Sweden

- **Konsult på Ericsson AB - Systems and Technology (R&D)**
  Plattformsforskning och utveckling av 5G- och 6G-testbäddar, från fältförsök till prototyper. Ansvarade för basbandsapplikationer på högre lager som körs på ett realtidsoperativsystem, med fokus på att minimera CPU-jitter och NIC-latens.
  - Utveckla en UDP-server och aktivera hårdvarutidsstämpling på NIC för att mäta paketens round trip time (RTT).
  - Utveckla en högpresterande AF_XDP-klient för paketbearbetning med zero-copy, för körning på Mellanox CX-6 200Gb/s och CX-7 400 Gb/s Ethernet nätverkskort som stöder XDP.
  - Utveckla ett testramverk och testfall i Python, Pexpect och YAML.
  - Skapa och konfiguera realtids OS baserat på Ubuntu. Supporta avdelningen och se till att realtidsapplikationen för basbandet kan köra på OS:et med höga krav på låg CPU-jitter.
  - Installera och ansvara för 5G och 6G hårdvaran och delar av mjukvaran i labbet.
  - Implementera zero-touch-automation med Ansible på Ericssons egenutvecklade hårdvara samt standardhårdvara (COTS) som Supermicro- och Dell PowerEdge-servrar med BMC/IPMI/iDRAC samt Mellanox SN4700 400GbE-switchar med Cumulus. Automationen genomför fullständig OS-distribution på flera noder parallellt och konfigurerar BIOS, LDAP och PTP, installerar Linux eller Cumulus OS obevakat samt sätter upp realtidsegenskaper med låg CPU-jitter och låg NIC-latens.

Dec 01, 2016–
Sep 30, 2018

Senior Software Architect and Design, *Realtime Embedded AB*, Stockholm, Sweden

- **Consultant at NorthStar**
  Development of NorthStar Advanced Connected Energy (ACE) gateway smart battery system.
  - Implement external alarm handling for the battery system. Ensure that the software can trigger and control relays through the GPIO pins in the gateway when correct signaling through IPC messages are received.
  - Implement functional tests and libraries/packages using the 9pm test framework.
- **Consultant at Vanderbilt (former Bewator)**
  Development of Linux Kernel drivers for access control system called Omnis. The hardware used was called E100 which is developed on site and is similar to AT91SAM9x5-EK.
  - Write GPIO test application to control relays.
  - Write Atmel AT91 ADC test application to read voltage raw values.
  - Write Linux kernel and device tree patch to enable USB-B gadget and Ethernet over USB.
  - Write Linux kernel driver and application for 74VHC595 8-bit shift register with output latches in order to control via sysfs LEDs, input ports and RS485 card readers.
  - Write Linux kernel driver and test application for Magstripe (Clock/Data) and Wiegand (Data/Data) card readers.
  - Write a firmware upgrade/flash script to automatically update the system using the SD/MMC card during bootup.
  - Modify and patch the bootloader from Atmel called at91bootstrap in order to fix bugs and enable bootup of SD/MMC or NAND flash.

Jun 10, 2011–
Nov 23, 2016

Software Developer and Scrum Master, *Ericsson AB*, Stockholm, Sweden

Employed at Ericsson in Kista as a Software Developer and Scrum Master. I developed drivers for the radio hardware units used in the Radio Base Station 6000 series. In order to first boot and bring up the radio hardware, the different drivers for the many components on the board needs to be working and configured correctly. This is done by either creating the drivers from scratch using the components data sheet and the boards circuit diagram or simply by adapting and reusing existing drivers. In most cases we emulate the hardware first and then write the drivers, before the physical boards arrive from the factory, to save time and money.

- Development of radio software drivers for the Radio SW platforms and Radio Hardware emulation. Board bringup and integration of hardware products like Antenna Integrated Radio unit (AIR), Micro Radios, Next Generation Radios and the 5G Program. Other side tasks include splitting parts of the Radio SW code into modules, introducing Git, Gerrit, Jenkins, BitBake and Autotools.
- Short term assignment (3 months) at the hardware department in Ottawa as a software integrator for Platform 5.2 AIR 32 products.
- Act as a technical Scrum/Kanban Master to lead and coach a Cross-Functional Driver team of 10 people. Conduct daily scrum meetings, sprint retrospective meetings, keep statistics of flow, eliminate waste and help Product Owner.

Aug 25, 2008–
Jun 09, 2011

Software Developer, *SAAB Combitech*, Stockholm, Sweden

My first consultant job started at SAAB Combitech for Ericsson AB Linux Center in Älvsjö and Telefonplan. There, I developed a test framework in Perl which is used to run automated test cases. The test cases was written in Tcl/Expect but the framework can also execute test cases in Perl or Bash and also support NetConf. The test framework became a separate product and was used to test different Linux embedded operating systems by many Ericsson products at the department. I continued in the same department but as a designer where I began to develop Linux kernel patches and pushed upstream to Linus Torvalds.

- Develop drivers and Linux kernel patches for the Ericsson Linux distribution called ENUX, which is used to run a Virtual Machine to emulate APZ classic run time environment for PLEX based programs such as AXE.
- Designer and tester for other Ericsson platform products such as CPBS, GSDI and IS running on GEP board hardware.
- Framework developer for automated test cases.
- Develop automated test cases for Ericsson products Linux Open-source Telecom Cluster (LOTC) and SLES Virtual Platform (SVP) based on Xen.
- Open Multimedia Platform (OMP) customer forum support.
- Handle and solve trouble reports for all the products.

2006–2007

Research Assistant, *University of Toronto*, Toronto, Canada

I worked as a research assistant for professor J. Christopher Beck at the Department of Mechanical and Industrial Engineering. My stay there was funded by a scholarship from the University of Toronto (UofT) and Toronto Intelligent Decision Engineering Laboratory (TIDEL). During my research, I became extremely interested in operations research, optimization and solving NP-hard combinatorial problems with the help of distributed algorithms. Together with another PhD student, we published a paper[1] and the whole experience abroad gave me new friends for life, deepened my analytical thinking and broaden my knowledge in research and software development.

- Research in artificial intelligence and planning. Published paper: "Modelling Security Protocol Synthesis with AI Planning", UofT TIDEL, ON, Canada.
- Research in optimization, parallel and distributed algorithms and communications protocol. Published paper: "Solving Combinatorial Problems with Parallel Cooperative Solvers", Ninth International Workshop on Distributed Constraint Reasoning, Brown University Providence, Rhode Island, USA.
- MPI Cluster Computing, installation and configuration.

Miscellaneous

2005–2006

Committee Member, *Luleå University Academic Computer Society*, City, Luleå

Served 600 registered student members.

2004–2005 **Treasurer**, *Luleå University Academic Computer Society*, City, Luleå
Responsible for a budget of over 180 000 SEK.

1999–1999 **Military Service, KA1 Waxholm**, *Swedish Government*, Stockholm, Sweden
- The royal guards at the royal palace in Stockholm and security guard.
- Leadership experience and basic medical education.
- Handle technical equipment for advanced management.

Education

2001–2006 **M.Sc. Computer Science & Engineering**, *Luleå University of Technology*, Luleå, Sweden
Specialized in Applied Mathematics

Master thesis

title *A Parallel Tabu Search Algorithm for the Quadratic Assignment Problem*

supervisors Professor J. Christopher Beck - University of Toronto & Professor Inge Söderkvist - Luleå University of Technology

description Find optimal solutions for a quadratic assignment problem using the tabu search algorithm on a cluster.

1995–1998 **Natural Science Programme**, *Västergård High School*, Södertälje, Sweden

Skills

	Level	Skill	Years	Comment
Programming Languages:	■■■□□	Assembler	1	*Regular exposure during my career.*
	■■■■■	C/C++	20	*I have used C/C++ in most of the big Linux embedded projects, Linux applications, kernel modules and patches.*
	■■■■□	Haskell	2	*I learned recursive programming at university.*
	■■■■□	Java	5	*From different courses at university and throughout work.*
	■■■■□	Lisp/Scheme	3	*Did the SICP course at University and to configure my Emacs editor.*
	■■■■□	Matlab	5	*Used in mathematics courses at the university and while working at Ericsson research.*
	■■■■■	Perl	7	*Wrote a test framework at Ericsson Linux DC that executed Perl and Expect test cases.*
	■■■■■	Python	15	*I wrote a test framework and corresponding test cases. Python has also been used for misc scripting in bigger projects throughout the years.*
	■■■■□	Ruby	2	*Used to write test cases at Radio SW. Very nice and friendly scripting language with a lot of handy predefined functions.*
	■■■■■	Shell scripting	23	*Bash, sed, awk regular expressions used in most of my work projects and at home to quickly write scripts with minimum dependencies.*
	■■■■■	Tcl/Expect	13	*I sometimes use Expect for automation and test cases in different projects.*

DevOps/Tools:	■■■■■	Ansible	3	*I use Ansible, Jinja2, YAML and Redfish for zero-touch automation.*
	■■■■■	Buildroot	3	*Used to build embedded Linux OS.*
	■■■■□	Docker	3	*Used when isolating applications at Ericsson Testbed.*
	■■■■■	Gerrit	10	*Code reviewing tool used in most of my projects together with Git and Jenkins.*
	■■■■■	Git	13	*Simply the best version control system out there.*
	■■■■■	Jenkins	10	*Combined with Gerrit and Git, you get the best tool for continuous integration.*
	■■■□□	SVN	1	*Version control system that I used when working at Vanderbilt.*
	■■■■■	Vim	20	*My main editor used for coding and editing.*
	■■■■■	Yocto/BitBake/Poky	3	*Used to build embedded Linux OS.*
OS:	■■■■■	Linux	23	*Started with RedHat, Slackware, Linux From Scratch and settled for Ubuntu.*
	■■■□□	OSE	5	*The main OS in the old Ericsson Radio product.*
	■■■■□	Solaris	5	*Primary OS at the university.*
Methods	■■■■■	Kanban/Scrum	5	*Team leader for Radio Software Board team.*

Languages

Swedish	Native speaker	*Born and raised in Sweden*
English	Fluent	*Thanks to early training since primary school*
Aramaic	Fluent	*Ancient christian language spoken by my parents*
German	Fair	*Learned in school*

References

Will be supplied on request.

Publications

[1] Lei Duan, Samuel Gabrielsson, and J. Christopher Beck. Solving combinatorial problems with parallel cooperative solvers, 2007.

[2] Samuel Gabrielsson and J. Christopher Beck. Modelling security protocol synthesis with ai planning, 2006.